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RESEARCH ARTICLE

Problems With SHAP and LIME in Interpretable AI for Education: A Comparative Study of Post-Hoc Explanations and Neural-Symbolic Rule Extraction

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ABSTRACT Given that education is classified as a 'high-risk' domain under regulatory frameworks like the EU AI Act, ensuring accurate and trustworthy interpretability in educational AI applications is critical due to its profound impact on student learning and development. This study compares a knowledge-based artificial neural network (KBANN) with a conventional artificial neural network (ANN) in the context of educational predictive modeling, focusing on generalizability, interpretability, and the fidelity of post-hoc explanations. While both models demonstrate comparable predictive performance, KBANN uniquely integrates structured educational knowledge, aligning more closely with essential educational principles and causal relationships. Post-hoc explanation methods, such as Kernel SHAP, Permutation SHAP, and LIME, were applied to the ANN to interpret its decision-making process, revealing significant variability in the assessment of feature importance. Through simulations based on the extracted rules from KBANN, we further examined the fidelity and reliability of these methods. Our findings, from global feature importance and correlation analyses, showed that post-hoc methods often fail to accurately reflect the structured knowledge learned by the model, misattributing importance to less relevant features. This misalignment and the resultant discrepancies in feature interpretation raise concerns about the reliability of these explanations, suggesting they may not always provide a trustworthy or accurate basis for understanding the predictive models. Finally, KBANN's rule extraction achieved significantly lower computation times compared to post-hoc methods, highlighting its practical efficiency. These findings collectively underscore the limitations of post-hoc explanation methods in conveying the true reasons behind model predictions, urging caution among stakeholders and researchers when using these methods for interpretability in AI for education.

INDEX TERMS Learner modelling, personalized learning, AI for education, explainable AI, neural-symbolic computing, SHAP, LIME.

I. INTRODUCTION

Machine learning has demonstrated promising potential in optimizing education through personalized learning, predictive and prescriptive analytics, and targeted intervention. Notably, artificial neural networks (ANNs) are among the most popular and successful machine learning methods in educational applications, outperforming traditional models in tasks such as predicting student performance, detecting undesirable behaviors, and crafting personalized recommendations [1], [2], [3], [4]. This has led to a significant increase in the use of ANNs to model learners' knowledge,

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activities, preferences, and characteristics [5], [6], [7], [8]. However, despite their potential, the practical application of ANN-based methods is often hindered by their lack of interpretability [9], [10]. These models typically operate as 'black boxes,' delivering results without clear explanations. This opacity poses not only technological challenges but also raises ethical, societal, and legal concerns, particularly in education—a domain where interpretability is paramount [11], [12], [13]. Furthermore, recent regulatory initiatives, such as GDPR and the EU AI Act, have highlighted the need for explanations for any significant or legally related decisions, and have even classified education alongside healthcare as a 'high-risk' domain, underscoring its profound impact on students' educational outcomes and future prospects, as well as the potential risks of discrimination [14], [15].

Interpretability in machine learning involves elucidating model decisions in a manner comprehensible to humans, revealing the underlying reasoning process [16]. This study, following [17], uses the terms interpretable and explainable interchangeably, collectively referred to as interpretability, and differentiates these from the term explanation, which pertains to the explanation of individual predictions. Moreover, interpretability is used as umbrella term capturing ''extraction of relevant knowledge from a machine-learning model concerning relationships either contained in data or learned by the model'' [18]. This attribute is especially critical in education, where clarity and fairness are paramount. The lack of interpretability can undermine trust in AI applications among educators and students, impeding effective learning and obscuring valuable feedback opportunities [12], [19], [20]. Clear explanations from AI systems are essential not only for understanding correct answers but also for offering precise educational guidance, which enhances the learning experience and fosters critical thinking. Moreover, interpretability aids in detecting and correcting biases within data and models, promoting fairness and equal educational opportunities. Recognizing its importance, recent studies have adopted post-hoc explanation methods like LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Additive exPlanations) to boost interpretability in educational AI applications, shedding light on how specific features influence predictions [2], [21], [22], [23]. These techniques, inherently universal, can be applied to any machine learning model, trained without initial explainability requirements. They aim to create an explainable approximation of 'black box' models, producing results that mimic the original model's outputs, thereby clarifying how decisions are made. LIME and SHAP are among the prominent techniques employed to enhance interpretability in machine learning. LIME focuses on local interpretability by generating perturbations around a specific data point and using a simpler model to approximate how each feature influences predictions at that locality [24]. SHAP, based on Shapley values from cooperative game theory, attributes the impact of a prediction fairly among all features. It calculates the contribution of each feature by assessing its presence and absence across all possible combinations, providing detailed insights into feature importance [25].

Despite the advantages of LIME and SHAP, both methods face inherent limitations that can impact their efficacy and reliability. A significant issue is that these approximating models are only indirectly related to the original model, meaning they may not accurately explain why the original model reached a particular conclusion. This limitation is particularly pronounced with LIME, whose assumptions of local linearity and feature independence often restrict its applicability when dealing with complex or interdependent data features [26], [27], [28]. The method's reliance on defining 'locality' and fitting surrogate models can also lead to unstable and unreliable explanations [29], [30]. Similarly, SHAP, while valuable for its detailed contribution analysis, struggles with computational efficiency, especially outside of tree-based models [31], and also grapples with the issue of lack of precision mostly due to their restrictive hypothesis of local linearity and feature independence as with LIME [26], [32]. For instance, according to studies conducted by [33], these methods may explain predictions without including relevant features in the explanation, or generating unrealistic scenarios. A recent joint research conducted by Harvard, MIT, Drexel, and Carnegie Mellon University [34] revealed that state-of-the-art explanation methods (including SHAP and LIME) often disagree in terms of the explanations they output for various models such a neural networks. More critically, they concluded that machine learning practitioners tend to rely on such misleading explanations for critical decision-making in the real world.

For ANNs, which are foundational to advancing AI in educational settings, interpretability goes beyond the surface-level explanations provided by LIME and SHAP [9]. The hidden layers of neural networks, where complex features and concepts are learned, necessitate more sophisticated interpretation methods [35]. These layers often encode the subtle patterns and nuances that are crucial for understanding deep learning processes [36]. By developing and employing interpretation methods tailored for neural networks, educators and technologists can uncover and understand the ''why'' behind predictions, not just the ''what.'' Building on the challenge of enhancing interpretability in ANNs, the integration of neural-symbolic AI presents a promising solution to bridge the gap between high performance and transparency [12], [37]. Neural-symbolic AI combines the robust predictive power of connectionist models (like ANNs) with the transparent, rule-based reasoning of symbolic AI. This hybrid approach leverages the strengths of both paradigms to provide deeper insights into the decision-making processes of machine learning models [35]. By incorporating a knowledge base that allows the system to learn and reason with symbolic representations, neural-symbolic AI not only enhances the production of understandable explanations but

also ensures that these explanations reflect meaningful causal relationships [9], [36]. This method supports the detection and correction of biases in training data, improves the model's robustness, and ensures that only relevant variables influence the outcomes [10], [36], [38]. Thus, neural-symbolic AI not only addresses the interpretability challenges faced by traditional neural networks but also advances the capability of AI systems to make fair, reliable, and transparent decisions in educational contexts.

This novel study in AI for education seeks to critically evaluate the effectiveness of neural-symbolic AI method, specifically the well-known knowledge-based artificial neural network (KBANN) method [39], in providing meaningful and interpretable explanations compared to established post-hoc explanation methods like SHAP and LIME. By integrating both training data and explicit rule-based reasoning, we trained the KBANN and compared its performance to an ANN trained solely on data. We evaluated both models using accuracy, precision, and recall to determine their predictive power. To better understand decision-making processes, we extracted rules from the KBANN and compared them with SHAP and LIME explanations for the ANN, and also compared SHAP and LIME explanations with each other. We then created wrapper models using these explanations and compared their performance against a reference model derived from KBANN's rules. This approach allowed us to quantitatively assess how closely SHAP and LIME explanations align with the rule-based reasoning of KBANN, thereby highlighting the effectiveness of these interpretability methods in replicating the reliability of rule-based models. We also compared the computation time of each method to evaluate their efficiency. This comparative study is crucial for advancing AI for education, as it not only tests the validity and efficiency of interpretability methods but also underscores the importance of neural-symbolic computing to achieve greater transparency and reliability in AI-driven educational tools.

II. RELATED RESEARCH

A. INTERPRETABILITY IN MACHINE LEARNING

Machine learning holds considerable promise across various fields, including education [2], [4], [7], [40]. However, the complexity of ML models often makes them difficult to interpret fully, posing challenges in understanding the decision-making process behind predictions [11], [19].

Interpretability in machine learning, as defined by [16], is "the degree to which a human can understand the cause of a decision." Higher interpretability means it is easier for someone to comprehend why certain decisions or predictions were made. Interpretable machine learning involves extracting relevant knowledge from a model about the relationships within the data or learned by the model. This is essential because a single metric, like classification accuracy, does not fully describe real-world tasks [41]. Understanding why a model makes certain predictions is crucial, especially in high-stakes scenarios like healthcare or self-driving cars. Interpretability helps detect biases, ensures fairness, reliability, and trust, and aids in debugging models. However, it is less critical in low-impact situations or well-studied applications, and sometimes, interpretability can lead to system manipulation. Ultimately, balancing predictive performance with the need for explanations depends on the context and consequences of the model's decisions.

Methods for machine learning interpretability are classified based on whether they are intrinsic or post hoc, the type of results they produce, and their model specificity. Intrinsic interpretability is achieved by using simpler models like decision trees and rule induction, while post hoc methods, such as permutation feature importance, analyze models after training. Results from interpretation methods can include feature summary statistics, visualizations, or insights into model internals like weights [24], [25]. Interpretations can explain individual predictions (local) or overall model behavior (global) [2], [23]. The scope of interpretability in machine learning can be classified into several levels [16], [17]. Algorithm transparency refers to understanding how the algorithm learns a model, independent of the data or specific learned model. Global, holistic model interpretability involves comprehending the entire trained model and its decision-making process. Local interpretability examines why a model made a specific prediction for an individual instance, offering potentially more accurate insights. Finally, local interpretability for a group of predictions combines methods for individual and global interpretations, to explain predictions for multiple instances [42].

In the educational context, there is no consensus on what interpretability in machine learning entails or how to measure it. Most existing works based on machine learning models that are not inherently interpretable employ post-hoc explanation methods to provide local and global explanations. For instance, [23] developed a learning analytics approach for student agency, using interpretability techniques to help educators understand the decision-making process. They demonstrated how SHAP values and LIME can provide both local and global explanations (feature importance) for their predictive models. Similarly, [22] put forward an interpretable student dropout prediction approach that uses post-hoc methods like permutation importance, LIME, and SHAP values to explain predictions made by models like the CatBoost classifier, providing educators with a deeper understanding of the factors influencing dropout rates and helping to design targeted interventions. Reference [43] presented an approach for predicting students' performance using machine learning models such as Logistic Regression. They conducted qualitative research with educational stakeholders to select practical features, and SHAP was used to visually provide information to help each student, demonstrating the potential for early identification of at-risk students and tailored educational support (for review of explainable AI in education, see [44] and [45]). Most of these studies mainly rely on

explanations provided by post-hoc explainer methods like SHAP and LIME, without critically comparing them or taking into account their inherent limitations and problems [30], [32], [33], [34]. For instance, [33] conducted a comparative study that revealed how high-fidelity explanations generated by post-hoc methods, despite their accuracy in mimicking black box predictions, can still mislead users by not accurately reflecting model biases.

B. INTERPRETABILITY IN ARTIFICIAL NEURAL NETWORKS FOR EDUCATION

The utilization of artificial neural networks (ANNs) in education has grown significantly. For example, a study by [46] explores the use of recurrent neural networks (RNNs) in knowledge tracing, significantly improving prediction performance on various datasets and aiding in intelligent curriculum design, indicating a promising new direction for RNN applications in educational settings. Reference [47] emphasizes the need for consistent training and testing of MOOC dropout predictors to avoid overly optimistic accuracy estimates, finding that deep, fully-connected feed-forward neural networks significantly improve test accuracy over logistic regression and perform competitively with post-hoc training. Reference [8] introduces an adaptive stacked denoising autoencoder for cross-session EEG-based mental workload classification, showcasing its superior performance in tracking students' mental workload and generalization capabilities for online implementation. Reference [5] demonstrates that deep learning techniques, specifically convolutional neural networks (CNNs), can effectively recognize students' emotions, thereby enabling educators to tailor their presentations based on students' emotional responses. Reference [6] presents DeepStealth, a deep learning-based stealth assessment framework utilizing LSTMs and n-gram encoded feed-forward neural networks, which reduces the labor of feature engineering and outperforms baseline models in predictive accuracy and early prediction capacity in game-based learning environments. Reference [7] investigates the effectiveness of transfer learning from deep neural networks for predicting students' performance in higher education, demonstrating that accurate predictions can be made using data from related courses. References [2] and [36] propose a deep knowledge tracing approach and a novel image-based learner modelling method using transfer learning, employing CNNs to model learners' knowledge state during gameplay, achieving high accuracy in predicting performance and solution correctness.

Although deep neural networks have demonstrated superior performance over traditional methods, significant challenges remain, particularly in providing interpretable predictions. Making predictions with a neural network involves passing data inputs through many layers of multiplications with learned weights and non-linear transformations. A single prediction can involve millions of mathematical operations, depending on the network's architecture. To interpret the behavior and predictions of neural networks, we need specific interpretation methods.

Post-hoc explainer methods have also been employed to provide interpretation in ANNs in educational contexts. For instance, [48] proposed a layer-based approach for student modelling and prediction of failure risk in online learning environments. They employed the LIME method to provide local explanations on the predictions made by deep learning in order to offer insight into the specific factors contributing to each student's predicted risk, thereby enhancing the transparency and interpretability of their predictive models. Reference [49] developed an approach for scoring essays using neural networks. Their use of SHAP implementations provided fast explanations of the model's decisions, thereby improving the interpretability of the proposed approach. The image-based learner modelling method proposed by [50] employed Grad-CAM to provide interpretability by highlighting the regions in the input images that contributed most to the predictions.

Aside from the inherent limitations and problems with such explainer methods (e.g., [26], [27], [28], [29], [30], [31], [32], [33], [34]), there are two reasons to consider interpretation methods developed specifically for neural networks: first, neural networks learn features and concepts in their hidden layers, necessitating special tools to uncover them; second, the gradient can be utilized to implement interpretation methods that are more computationally efficient than model-agnostic methods that analyze the model from the outside.

In a different approach, [51] proposed using the post-hoc method to tackle the interpretability issue for deep learning-based knowledge tracing models. Specifically, they applied the layer-wise relevance propagation (LRP) method to interpret RNN-based deep learning-based knowledge tracing models by backpropagating the relevance from the model's output layer to its input layer. Their experiments demonstrated the feasibility of using the LRP method for interpreting deep knowledge tracing model predictions, showing that the method could provide insight into the model's decision-making process and validate the relevance scores for practical applications. While useful, this method cannot provide an explanation about the inner mechanism of the model. In the educational context, understanding the latent features and concepts that neural networks learn is particularly important because human learning is often about uncovering and mastering these latent patterns. For example, identifying underlying factors that contribute to a student's success or struggles can help educators tailor interventions more effectively. In a recent attempt, [36] introduced a neural-symbolic AI approach that integrates neural networks with knowledge representation and symbolic reasoning to enhance interpretability and rule extraction in educational applications. The approach allows for the extraction of rules from the trained network, providing clear reasoning for the decision-making process and

uncovering the features and concepts learned in the hidden layers.

III. METHOD

This study compares the performance and interpretability of the KBANN method and a conventional ANN. The methodology includes detailed data preparation, model implementation, and the use of various explanation methods, such as SHAP with Kernel and Permutation explainers, and LIME. The evaluation framework assesses fidelity (alignment between post-hoc explanations and KBANN predictions) and reliability (trustworthiness and consistency of feature importance explanations and KBANN rule explanations) through correlation analysis and classification metrics, while also comparing computation times to provide a comprehensive assessment of performance and efficiency.

A. THE AutoThinking GAME

AutoThinking is an educational game aimed at enhancing computational thinking (CT) by using icons instead of traditional programming syntax, thereby preventing syntax errors. As one of the pioneering adaptive games in this domain, AutoThinking incorporates adaptiveness in both its gameplay and educational processes. Players control a mouse navigating through a maze to gather cheese, avoid cats, and accumulate points. The game emphasizes four critical CT skills: breaking down problems (algorithmic thinking), building algorithms (pattern recognition and generalization), debugging, and simulating solutions. It also introduces essential programming concepts such as action sequencing, conditional decision-making, and looping. AutoThinking allows players to devise up to 20 strategies to collect 76 cheese pieces, providing adaptive feedback and hints to support effective problem-solving and algorithmic thinking. Fig. 1 shows the user interface of the third level of the AutoThinking game.

B. THE PROPOSED APPROACH

Fig. 2 and Table 1 provide an overall architecture and algorithm of the proposed comparative approach. The approach involves several key steps including data preparation, model implementation, explanation methods, and simulated SHAP- and LIME-based predictions, as well as their comparative analysis.

1) DATA PREPARATION

We acquired data for this research using a customized function designed to process and read structured csv files. This function systematically extracts and segregates features and labels, storing them into distinct data structures for further analytical processing. We initially disregard the output variable column to ensure that the analysis focuses solely on the relevant input variables necessary for model training and evaluation. During the pre-processing phase, we transformed the data structures to a uniform format to ensure compatibility with the analytical models used later in the study. More detailed information on the data description can be found in section ''Datasets and Educational Knowledge.''

2) MODEL DEVELOPMENT

a: KNOWLEDGE-BASED ARTIFICIAL NEURAL NETWORK (KBANN)

In this study, we used a neural-symbolic AI (NSAI) framework to incorporate educational knowledge into the neural network's design [39]. This approach helps the network not only learn from data but also provide explanations for its decisions that are rooted in domain-specific knowledge. The process begins by loading predefined symbolic rules, which are simplified and adapted to fit into the neural network's structure. These rules guide the configuration of the network, determining how layers, weights, and connections are set up. As the network is trained, a modified backpropagation method is used to ensure that the learning process aligns with the integrated educational knowledge. This helps the network adjust its parameters while maintaining the integrity of the symbolic rules. After training, the network generates explanations by extracting and reconstructing rules that explain its predictions, making the model's reasoning more understandable to humans.

b: ARTIFICIAL NEURAL NETWORK (ANN)

We implemented a conventional ANN model using the Keras library, which included a sequence of dense layers optimized using Adam optimizer [52]. The ANN was trained on the feature data, with the model architecture comprising input, hidden, and output layers designed to capture the complex patterns in the data without the explicit rule-based structure used in KBANN. Section ''Experiment Setting'' offers more information on the structure and parameters of the models.

3) EXPLANATION METHODS

In the selection of explanation methods for this study, we considered the unique nature of our model, which primarily operates based on rules extracted from the KBANN model, and included both SHAP [25] and LIME [53] to provide comprehensive explanations. We chose to use SHAP values limited to Kernel and Permutation explainers due to their model-agnostic capabilities. While other SHAP explainers like TreeExplainer, DeepExplainer, and GradientExplainer offer valuable insights for specific model types such as tree-based, deep learning, and linear models, we did not employ them as our models require a flexible approach that does not presume the presence of gradients or decision paths, making Kernel and Permutation SHAP explainers the most suitable tools for providing meaningful and accurate explanations.

a: SHAPLEY ADDITIVE EXPLANATIONS (SHAP)

SHAP values were computed using three different explainers: Kernel, Permutation, and the integrated SHAP explainer.

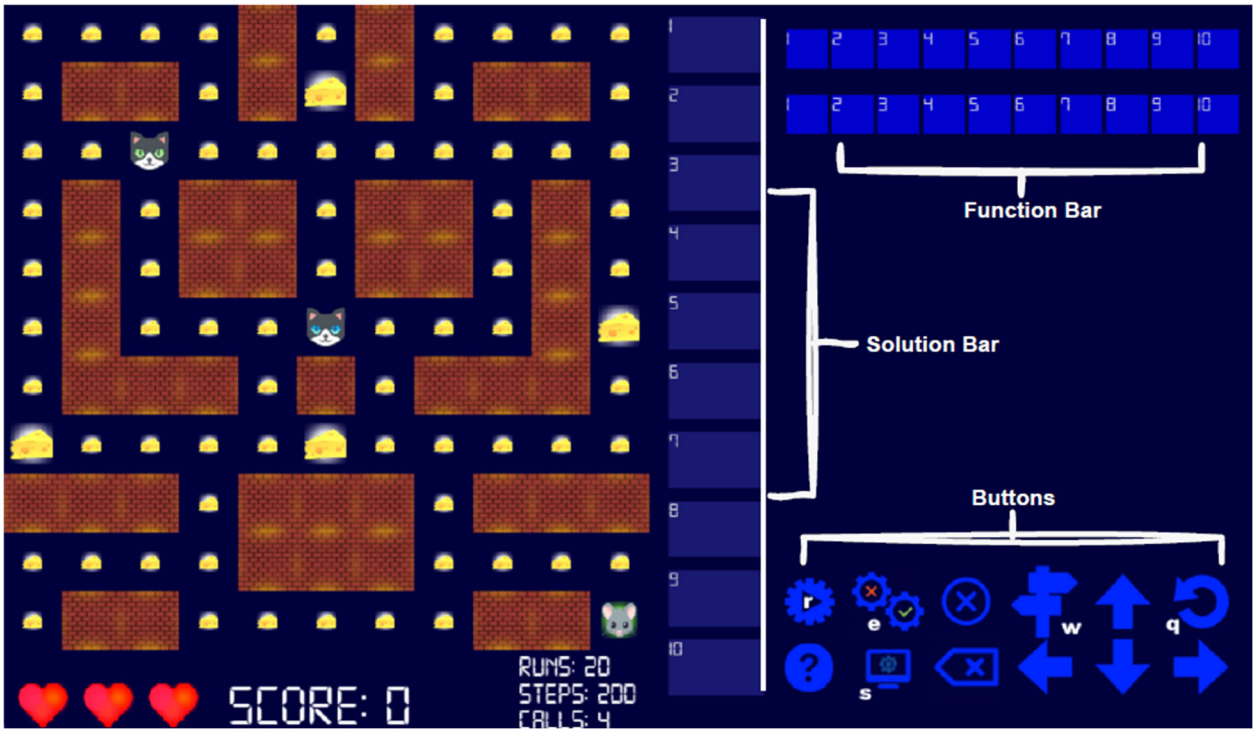


FIGURE 1. Interface of the AutoThinking game.

These explainers provided insights into the contribution of each feature to the predictions made by the model. For the explainers, the background dataset needed for comparison was based on the training data, ensuring the explanations reflected the general behavior of the model as observed from typical instances.

Kernel SHAP explainer utilized a model-agnostic approach, applying a weighted linear regression to estimate the impact of each feature. In contrast, Permutation explainer assessed feature importance by evaluating the change in model accuracy resulting from the permutation of feature values, thereby estimating their impact on the model's predictions. The integrated SHAP explainer, directly interacting with the model's prediction function, used a background dataset to compute SHAP values, offering an explanation leveraging the model's internal mechanisms. Together, these explainers provided a comprehensive view of feature importance, allowing for a robust analysis of the factors driving the model's predictions.

b: LOCAL INTERPRETABLE MODEL-AGNOSTIC EXPLANATIONS (LIME)

LIME explanations were generated for each instance in the test dataset, highlighting the local feature contributions towards the prediction of each instance. This was achieved through a linear model, fitted locally around each prediction, which provided a straightforward interpretation of feature impact by using perturbed samples around the data point of interest.

4) SIMULATED SHAP- AND LIME-BASED PREDICTIONS

In our simulated prediction processes for SHAP (Kernel and Permutation) and LIME, we apply distinct methodologies to predict outcomes based on the interpreted feature contributions, each reflecting a unique approach to interacting with the KBANN model. First, we created a wrapper model from KBANN. Then, we implemented SHAP and LIME on the wrapper to simulate, assessing the extent to which they can replicate KBANN's predictions.

For the Kernel SHAP explainer, we simulate predictions by summing the SHAP values for each feature per test instance and then comparing this sum to a baseline value known as the expected value. This expected value represents the average output across the background dataset used for the explanation and serves as a threshold. If the sum of SHAP values plus the expected value exceeds this threshold, the prediction is classified as class 1; otherwise, it is class 0. This approach allows us to integrate the overall impact of features while accounting for the model's base prediction tendency when no specific feature information is given. In contrast, the Permutation SHAP explainer does not utilize an expected value but instead directly aggregates the SHAP values. Here, the decision threshold is zero, where a positive sum indicates a prediction of class 1, and a sum less than or equal to zero suggests class 0. This method emphasizes the net effect of feature permutations on the model's accuracy, focusing on the incremental importance of each feature in isolation. For LIME, the total influence of features, represented by the sum of their contributions as determined by LIME, is compared

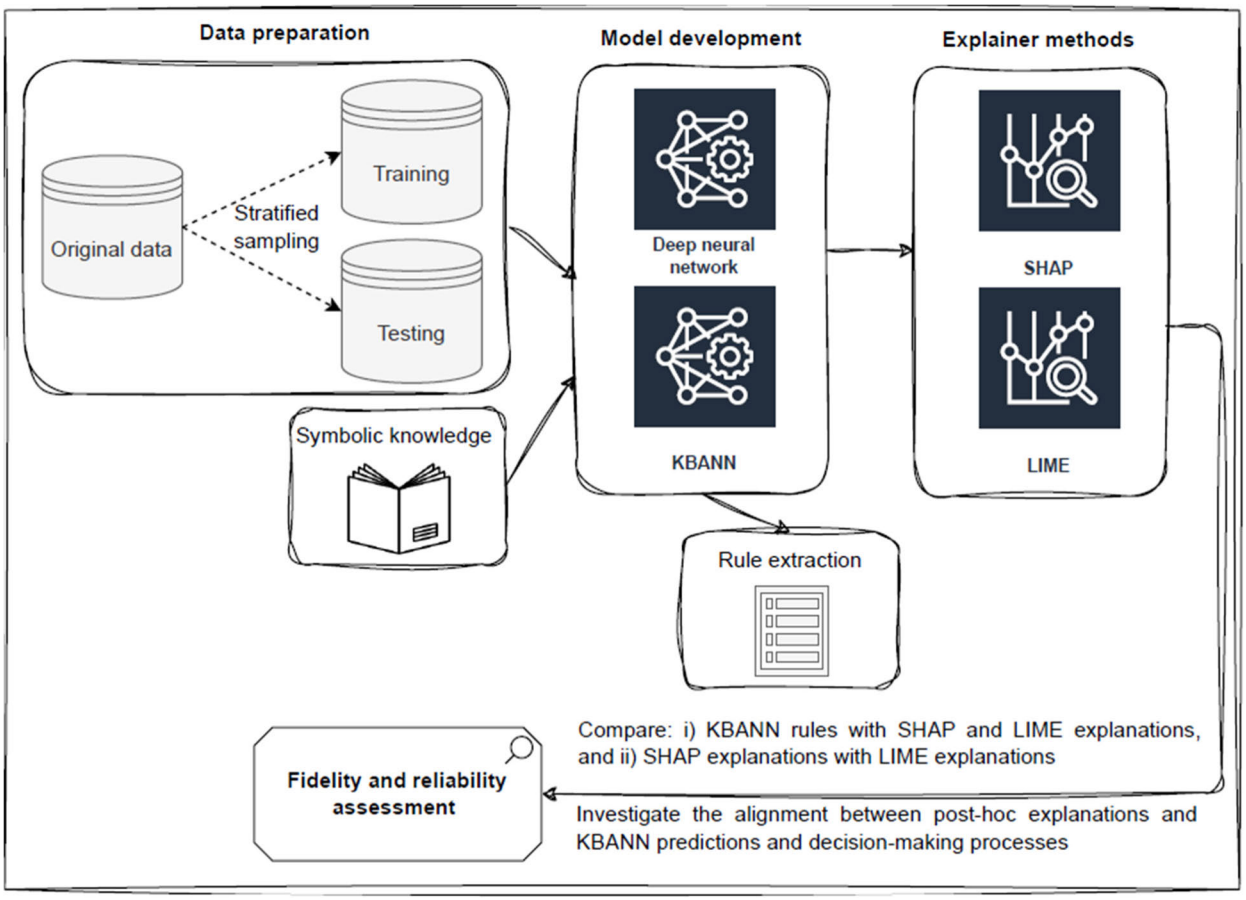


FIGURE 2. Overall architecture of the proposed methodological framework.

against a zero threshold. A positive total results in a prediction of class 1, while a negative or zero total leads to a class 0 prediction. This approach offers insights into individual prediction dynamics, highlighting which specific features most significantly influence the outcome in a localized context.

5) COMPARATIVE ANALYSIS AND RELIABILITY ASSESSMENT

In our comparative analysis, we first developed both KBANN and ANN models, comparing the KBANN rules with ANN explanations using SHAP and LIME, and employing metrics such as accuracy, precision, recall, and F1-score. Accuracy provides a general measure of overall performance, while precision and recall offer insights into the models' ability to correctly classify instances of each class. The F1-score, the harmonic mean of precision and recall, balances the trade-offs between these metrics. We also quantitatively compared the SHAP and LIME explanations with each other using feature importance scores from all four different model explainers. We then created a wrapper model from KBANN rules to generate predictions, simulating and evaluating these explainer models' predictions against the KBANN rule-based predictions on test data using the same metrics of accuracy, precision, recall, and F1-score, as well as correlation analysis. Additionally, we compared SHAP and LIME explanations with each other and assessed their fidelity by examining how well they matched the KBANN model's predictions and feature importance distributions. We also conducted a detailed reliability assessment by comparing the normalized average absolute values of explanations to those derived from the model's intrinsic rules. This comprehensive evaluation framework illuminated the alignment and divergence between post-hoc explanations and the underlying rules of the KBANN model, providing insights into the consistency, trustworthiness, and visual representation of feature contributions and model behavior.

IV. RESULTS AND ANALYSIS

A. DATASETS AND EDUCATIONAL KNOWLEDGE

To predict learners' solution scores in the CT game of AutoThinking, we analyzed data from 442 players who participated in the third level of the game. The system recorded various learner interactions during gameplay, such as tracking player and non-player characters' positions, task identifiers, and the collection of small and large cheeses. Additionally, it monitored the usage of loops, conditionals, arrows, and functions, as well as debugging and simulation activities. The system also tracked the frequency of seeking help, the amount

TABLE 1. Overall algorithm of the proposed comparative approach.

Step	Description
1	Load data and symbolic knowledge: • Load and parse training and test datasets. • Load and parse rules.
2	Train KBANN model: • Integrate symbolic rules with the neural network architecture. • Train the KBANN model using the training data. • Extract rules from the trained KBANN network.
3	Calculate feature importance for KBANN: • Parse extracted rules. • Calculate and normalize feature importance from the KBANN rules.
4	Train ANN model: • Define, compile, and train the ANN model on the training dataset.
5	Evaluate models: • Predict and evaluate labels for the test dataset using both KBANN and ANN models. • Calculate performance metrics (accuracy, precision, recall, F1-score) for both models.
6	Explain ANN model with SHAP and LIME: • SHAP KernelExplainer: Compute and normalize SHAP values for the test dataset. • SHAP PermutationExplainer: Compute and normalize SHAP values for the test dataset. • LIME: Generate explanations for instances in the test set, aggregate, and normalize feature importance values.
7	Simulate KBANN predictions and explanations: • Define a wrapper model using KBANN rules. • Generate predictions for the test dataset using the KBANN rule wrapper. • Simulated SHAP-based predictions: ○ Kernel SHAP: Sum SHAP values for each test instance and compare this sum to the expected value. Classify as class 1 if the sum plus the expected value exceeds the threshold; otherwise, classify as class 0. ○ Permutation SHAP: Aggregate SHAP values directly. Classify as class 1 if the sum is positive; otherwise, classify as class 0. • Simulated LIME-based predictions: ○ Sum the contributions of features as determined by LIME. Classify as class 1 if the total influence is positive; otherwise, classify as class 0. • Evaluate the simulated predictions using performance metrics.
8	Compare feature importance: • Normalize and compare feature importance from KBANN, SHAP (Kernel and Permutation), and LIME. • Visualize the comparisons using bar plots.
9	Calculate Pearson correlation coefficients: • Compute Pearson correlation coefficients between KBANN feature importance and those from SHAP and LIME.
10	Comparing computation time • Compute the computation time of rule extraction from KBANN and compare it with that of SHAP and LIME.

of feedback and hints received, the frequency of collisions with walls, the estimation of learners' CT knowledge, and the evaluation of solution quality based on a Bayesian network [54]. For this research, we focused on features that were directly or indirectly related to players' solution scores.

TABLE 2. Descriptive statistics of the original dataset.

Features	Mean	std	min	max
Arrow	5,7	3,16	0	10
Loop	0,45	0,78	0	3
Conditional	0,2	0,58	0	3
Function	0,04	0,24	0	4
Debug	0,06	0,23	0	1
Simulation	0,2	0,4	0	1
HitWall	0,43	1,5	0	14
SmallCheese	4,38	3,98	0	33
CrazyCheese	0,11	0,32	0	2
BigCheese	0,12	0,32	0	2
Solution Score	0,32	0,47	0	1
	Low= 4401		High= 2035	

Table 2 shows the descriptive statistics of various features in the dataset. The features "Arrow," "SmallCheese," and "HitWall" have the highest means, indicating they occur more frequently in the data. The standard deviations show significant variability in "Arrow" and "SmallCheese" compared to other features. Most features have a minimum value of 0, with maximum values ranging significantly, especially "SmallCheese" with a maximum of 33. The "Solution Score" has a mean of 0.32, suggesting a generally low scoring trend across the dataset. Additionally, the dataset contains 4401 false instances and 2035 true instances, highlighting an imbalanced distribution. Fig. 3a to 3c uses deviation charts to illustrate the distribution of the main dataset before splitting, and after splitting using stratified sampling of 80%-20% for training and test set.

Fig. 4 illustrates the correlation analysis to further show the relationship amongst the input features and the label. The "Loop" feature has the strongest positive correlation with the solution score (0.58), indicating that proficiency in using loops significantly enhances performance. "SmallCheese" (0.40) and "Conditional" (0.34) also show high positive correlations, highlighting their importance in achieving higher solution scores. "Simulation" (0.29) and "Function" (0.18) are moderately correlated, while "CrazyCheese," "BigCheese," and "Debug" (all around 0.14-0.15) have weaker positive correlations. Conversely, "HitWall" (−0.10) and "Arrow" (−0.24) exhibit negative correlations, suggesting that higher occurrences of these features are associated with lower solution scores. This analysis underscores the importance of loop and conditional logic in scoring well,

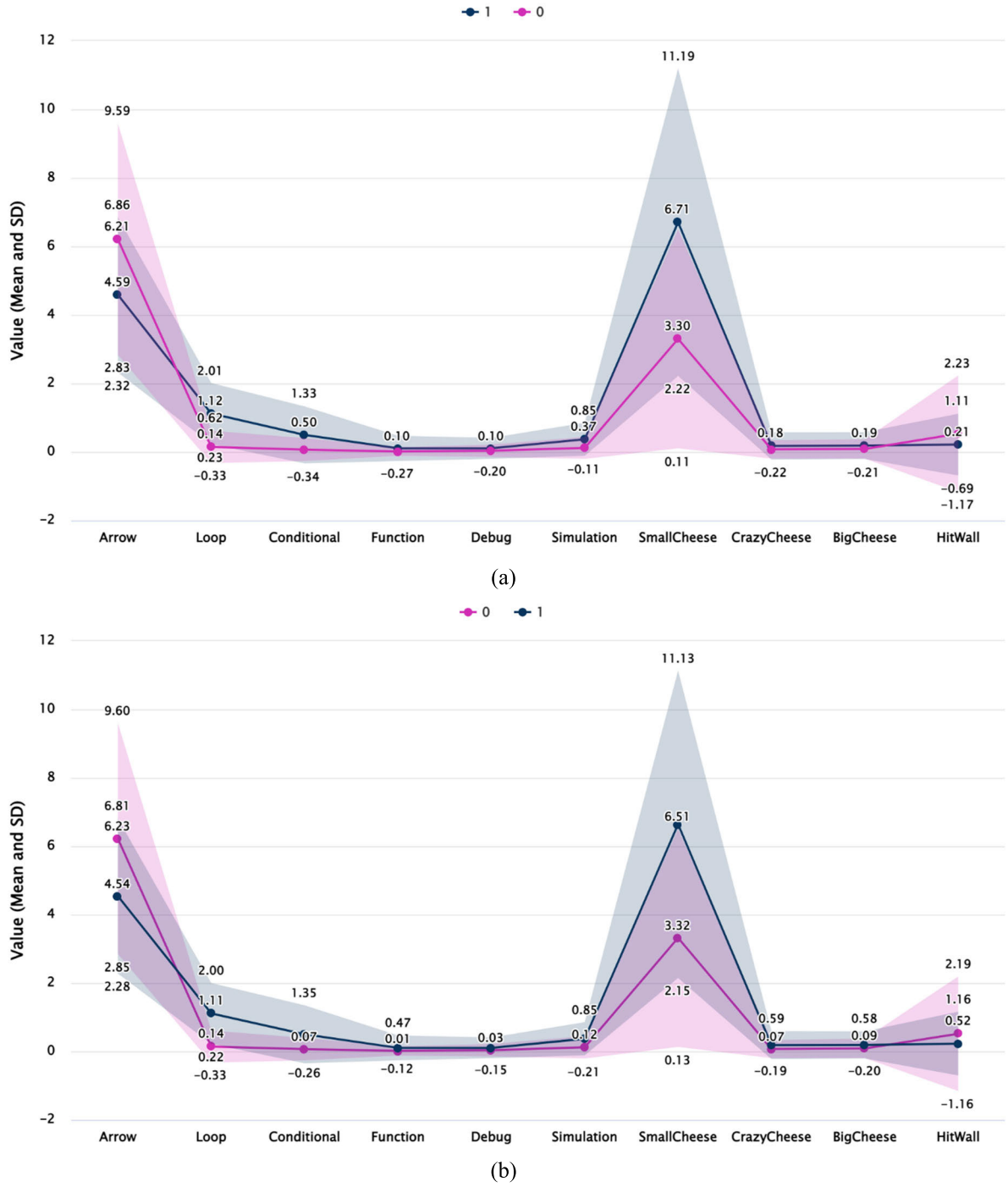


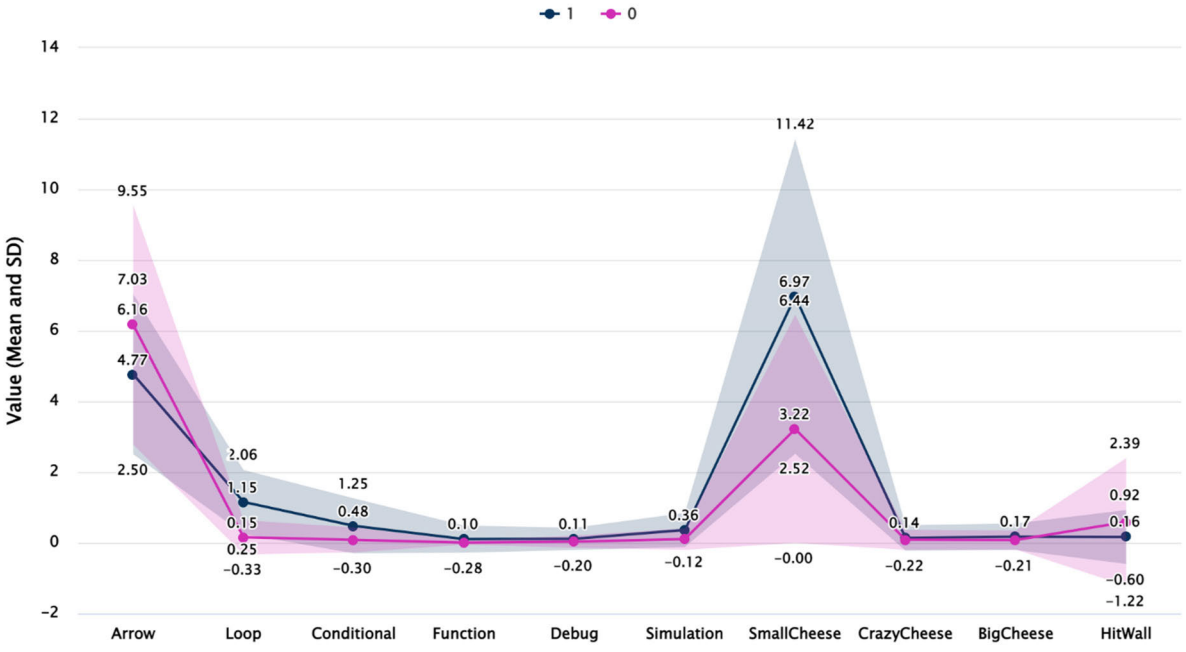
FIGURE 3. Deviation chart to visualize the distribution of the: a) original datasets before splitting, b) training data.

while the incorrect use of arrows and hitting walls negatively impacts performance.

We utilized the CT skills and concepts from [12] to integrate domain (symbolic) knowledge into the NSAI method. Briefly, it entails skills like identifying problems, creating algorithms, debugging, and running simulations, along with concepts such as sequences, loops, and conditionals. We represented this knowledge in the neural networks using propositional logic as follows:

- SolutionScore: CT_concepts, CT_skills.
- CT_concepts: Conditional, Loop, Arrow.
- CT_skills: Debug, Simulation, Function, HitWall.

In short, the first logic rule shows that a learner's final score or performance in the game relies on their mastery of both CT



(c)

FIGURE 3. *(Continued.)* **Deviation chart to visualize the distribution of the: test data.**

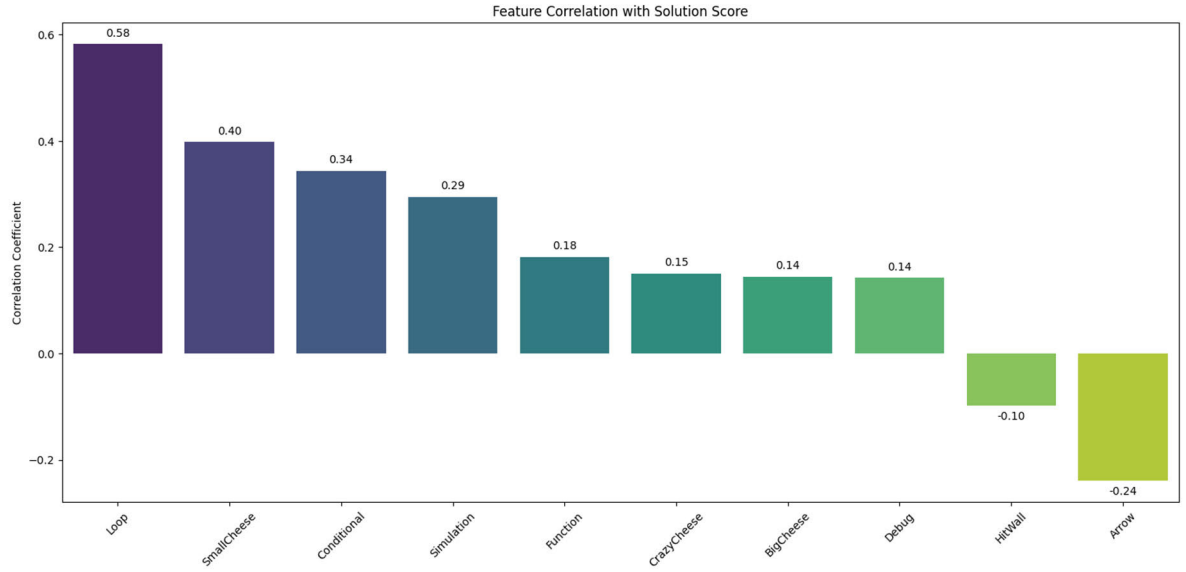


FIGURE 4. **Ranking of feature correlations with class label (Solution score).**

skills and concepts. It further explains that CT concepts are tied to features like conditional statements, loops, and arrow usage in the dataset, while CT skills are linked to debugging, simulation, function usage, and avoiding hitting walls. Good CT skills help learners debug effectively, simulate accurately, and use functions properly, reducing the chances of hitting walls, which symbolizes avoiding errors or obstacles. Essentially, learners perform well in CT if they excel in both CT concepts and skills.

B. EXPERIMENT SETTING

The computer utilized for our implementation featured a 12th Gen Intel(R) Core (TM) i9-12900K CPU with 128.0 GB of memory. The deep learning model was structured as a multilayer, feed-forward artificial neural network, which underwent training via Adam optimizer (learning rate of 0.001). The model consisted of two fully connected layers with ReLU activation functions and 32 neurons each, followed by an output layer with a sigmoid activation function

and one neuron. The KBANN model employs both fully connected layers and additional hidden units (representing the symbolic knowledge), ensuring a comprehensive learning process. Sigmoid activation functions are employed throughout the network to compute activations. In our experiments, we set the learning rate to 0.001 and trained the model, monitoring performance every 1,000 steps. During training, the network weights and biases were refined, and regularization techniques, such as dropout, were employed to improve generalization and prevent overfitting.

Regarding the explainer methods, we employed SHAP explainer, Kernel SHAP, Permutation SHAP, and LIME to interpret the ANN and KBANN models' predictions. For Kernel SHAP, we used a background dataset sampled from the training data to compute SHAP values, setting the sample size to 100 to balance between computational efficiency and explanation precision. Permutation SHAP was configured to measure feature importance by permuting the entire test dataset, directly evaluating the impact of feature shuffles on model predictions. The integrated SHAP explainer also utilized a background dataset sampled to 100 to provide comprehensive SHAP values for the ANN model, but it was not used for the KBANN model wrapper (experiment 2) since Kernel SHAP gave very similar results in our experiments with the ANN (see Table 5). LIME was configured with LimeTabularExplainer, using the entire training set as the reference for discretizing continuous features.

C. EXPERIMENT 1: COMPARING KBANN PERFORMANCE WITH ANN IN TERMS OF GENERALIZABILITY AND INTERPRETABILITY

Table 3 compares the performance of KBANN and the ANN in terms of generalizability across various classification metrics. Both models achieve the same overall accuracy of 0.90. The precision, recall, and F1-score for class Low and class High are also similar between the two models. Specifically, the precision for class Low and class High is 0.92 and 0.87, respectively, for both models. The recall for class Low is slightly higher in ANN (0.82) compared to KBANN (0.81), while the recall for class High is identical at 0.94. The F1-score is also similar. This indicates that both models perform comparably well in terms of generalizability.

TABLE 3. Performance of the KBANN and ANN on the test data.

Models	Accuracy	Precision		Recall		F1-score	
		Low	High	Low	High	Low	High
ANN	0.90	0.92	0.87	0.94	0.82	0.93	0.84
KBANN	0.90	0.92	0.87	0.94	0.81	0.93	0.84

1) EXPLANATION OF KBANN

The extracted rules from the NSAI approach provide a comprehensive understanding of how various features contribute to predicting learners' solution scores in CT (see Table 4). A high solution score is predicted when the combined weighted contributions of CT skills, CT concepts, and latent nodes (head1 and head2) exceed a specific threshold, such as the value 10.457. For example, if a learner's CT skills and CT concepts, along with their associated heads, produce a value greater than 10.457, the model predicts a high performance. This highlights that proficiency in both CT skills, like debugging and simulation, and CT concepts, such as conditionals and loops, is essential for achieving a high score.

CT skills and CT concepts as well as head1 and head2 are latent features, meaning they are not directly observable but are inferred through various observable actions and interactions during gameplay. For instance, CT concepts are influenced by features such as Conditional, Loop, and Arrow, while CT skills are impacted by Debug, Simulation, and Function. Head1 is mainly influenced by SmallCheese, Loop, Conditional, and Simulation, while head2 is influenced by Simulation, Conditional, Loop, and BigCheese. These latent features capture the deeper understanding and application of CT principles by the learners. Consequently, the NSAI model incorporates the causal relationships related to CT skills and concepts, presented as educational knowledge in the form of rules and accordingly learns the underlying patterns from the training data. This implies that, unlike the ANN model, which mostly focuses on small cheese consumption (which is considered spurious correlations; for more information, see [36]) to improve accuracy on the test data, the NSAI model incorporates educational knowledge and adheres to educational restrictions while learning from the data to improve test data accuracy. It is worth noting that the extracted rules also allow for the initial educational knowledge to be revisited using the training data. In this way, CT skills play a more important role in predicting the solution performance of learners compared to CT concepts, as indicated by the weights associated with CT skills and concepts in the rules.

2) EXPLANATION OF ANN

The bar chart depicted in Fig. 5 shows the SHAP explainer global feature importance plot for the ANN, providing a detailed visualization of how various features impact the model's output, quantified by their mean SHAP values. The x-axis represents the mean SHAP value, indicating the extent of each feature's influence on the model's predictions. Features such as "SmallCheese," "Arrow," and "Loop" are prominently displayed with the highest mean SHAP values of 0.16, 0.15, and 0.12, respectively, signifying their significant influence on the ANN's predictions. "SmallCheese" and "Arrow," in particular, exhibit the highest impact, indicating a consistently strong effect on the model's predictions. On the other hand, features like "Debug," "HitWall," and "Function" show minimal to no impact, with mean SHAP values of 0.01 or less, highlighting their lesser importance in the ANN model's predictive power.

Fig. 6a and 6b present SHAP force plots for the solution score labels Low and High, illustrating how different features influence the model's predictions towards these labels. For the Low label, "SmallCheese" and "CrazyCheese"

have substantial positive contributions, pushing the model's prediction towards label Low (0). In contrast, features like ''Arrow,'' ''Loop,'' ''Conditional,'' ''Simulation,'' and ''BigCheese'' have negative SHAP values, indicating they push the prediction away from label Low. For the High label, ''SmallCheese,'' ''Arrow,'' and ''Loop'' have substantial positive contributions, pushing the prediction towards label High (1). ''SmallCheese'' has the highest impact with a SHAP value of 11.0, followed by ''Arrow'' with 4.0 and ''Loop'' with 2.0. Although there are minimal blue segments indicating slight opposing contributions, the strong positive SHAP values of the key features demonstrate a strong inclination towards predicting label High. This combined analysis effectively highlights the key drivers influencing the model's predictions towards both labels.

Table 5 compares the feature importance scores from four different model explainers: Kernel SHAP, Permutation SHAP, LIME, and SHAP explainer. Across all explainers, the features ''Arrow'' and ''SmallCheese'' consistently emerge as the top factors, with all methods assigning them very high importance scores (close to 1). This agreement underscores the significance of these features in the model's predictions. Conversely, features such as ''Function,'' ''HitWall,'' and ''Debug'' are consistently rated as the least important across all methods, with low importance scores (around 0.01 for ''Function'' and 0.03 for ''Debug'' in Kernel SHAP and SHAP explainer). This consistency indicates a general consensus on their minimal impact on the model's output. However, notable differences are observed in the importance scores for features like ''Conditional'' and ''Simulation.'' For instance, LIME assigns a high importance score to ''Conditional'' (0.70) compared to the other explainers, which score it around 0.28. Similarly, ''Simulation'' receives a significantly higher score from LIME (0.12) than from the other methods. Furthermore, there are slight differences between Kernel SHAP and Permutation SHAP, such as for ''Simulation'' (0.053 vs. 0.056) and ''BigCheese'' (0.063 vs. 0.065), indicating that even closely related methods can yield subtly different insights. These discrepancies highlight the variability in feature importance interpretation depending on the explainer used. This comparison underscores the importance of using multiple explainers to gain a comprehensive understanding of model behavior, especially when there is no clear consensus on the importance of certain features.

The subsequent experiment seeks to evaluate how effectively these explanations replicate the inherent decision-making processes of the predictive model, and to assess whether they reliably reflect the true underlying mechanisms of predictive models.

D. EXPERIMENT 2: INVESTIGATING THE FIDELITY AND RELIABILITY OF THE SHAP AND LIME METHODS

1) PERFORMANCE OF THE SIMULATED PREDICTIONS AND CORRELATION ANALYSIS

Table 6 and 7 illustrate the performance results of the simulated predictions and correlation analysis. The analysis of simulated predictions using SHAP (Kernel and Permutation) and LIME provides insights into their performance metrics and correlations with KBANN's inherent decision-making processes. Kernel SHAP and LIME exhibit variations in performance, particularly in precision for class High and overall F1-score with Kernel SHAP, indicating discrepancies in how these models prioritize and interpret features compared to KBANN. LIME shows slightly better performance than Kernel SHAP but still falls short of matching KBANN's performance, particularly in recall for class Low and overall F1-score. While the Permutation SHAP method's metrics align closely with KBANN's in terms of accuracy, precision, recall, and F1-score, it is crucial to understand that this similarity in performance metrics may not necessarily reflect an accurate replication of KBANN's decision-making process.

The correlation analysis in Table 7 provides additional insight into how closely the predictions from these models align on a per-instance basis. According to the correlation analysis, there is a relatively low Pearson correlations between KBANN and all three post-hoc explanation models. LIME shows the highest correlation at 0.447216, while both SHAP Kernel and Permutation show only marginal correlations. These modest correlations indicate that despite recognizing some influential features consistent with KBANN's theoretical constructs, the explanations and feature importance interpretations offered by these models differ significantly from those of the reference model. This disparity underscores the limitations of post-hoc explanations in faithfully reproducing and interpreting the underlying reasons for predictions made by theoretically-informed models like KBANN.

2) GLOBAL FEATURE IMPORTANCE EXPLANATION

To further investigate the explanatory power and interpretive fidelity of the SHAP and LIME methods, Fig. 7, 8 and 9 illustrate Kernel and Permutation SHAP scatter and heatmap plots, as well as LIME bar chart. These figures allow to compare how effectively these methods can elucidate the feature-based dynamics influencing the model's predictions in comparison to the KBANN model.

The Kernel SHAP heatmap plot demonstrates the impact of various features on a model's output across more than 1200 instances. Each row in this plot corresponds to a distinct feature, such as ''SmallCheese,'' ''Loop,'' and ''Arrow,'' where the intensity and hue of the colors (blue for negative, red for positive) signify the magnitude and direction of the feature's influence on the model output. The horizontal spread of colors within each row shows the distribution of SHAP values for each feature across the instances, highlighting the variability in their significance. The vertical bar plot on the right side aggregates the average impact of each feature, providing a rapid assessment of which features predominantly influence the model's predictions. This plot not only reveals insights into the relative importance and contribution of each feature but also underscores the variability of these contributions in different predictive scenarios. The

TABLE 4. Knowledge extracted from the trained KBANN model to explain the predictions.

Variables	Rules
SolutionScore:	10.457 < 5.649 * (CT_skills) + 3.084 * (CT_concepts) + 4.801 * (head1+head2)
CT_concepts:	23.485 < 9.633 * (Conditional) + 0.134 * (CrazyCheese) + 2.281 * (SmallCheese) + -0.084 * (HitWall) + 9.837 * (Loop) + 9.529 * (Arrow) + 0.094 * (Simulation) + 0.162 * (Function+BigCheese) + 0.070 * (Debug)
CT_skills:	33.419 < 8.493 * (HitWall) + 0.028 * (CrazyCheese) + 9.539 * (Debug+Simulation) + -1.763 * (Arrow) + 0.326 * (SmallCheese) + -0.076 * (Conditional) + 9.628 * (Function) + 0.068 * (BigCheese) + 0.003 * (Loop)
head1:	1.611 < 0.488 * (SmallCheese+BigCheese+CrazyCheese) + 1.975 * (Loop) + -0.537 * (Arrow) + 1.478 * (Conditional) + 0.289 * (Debug) + -0.703 * (HitWall) + 0.530 * (Simulation+Function)
head2:	0.295 < 0.626 * (Simulation) + 1.637 * (Conditional) + -0.248 * (Arrow) + 0.231 * (Debug+SmallCheese) + 2.163 * (Loop) + -0.676 * (HitWall) + 0.519 * (CrazyCheese) + 0.429 * (BigCheese) + 0.607 * (Function)

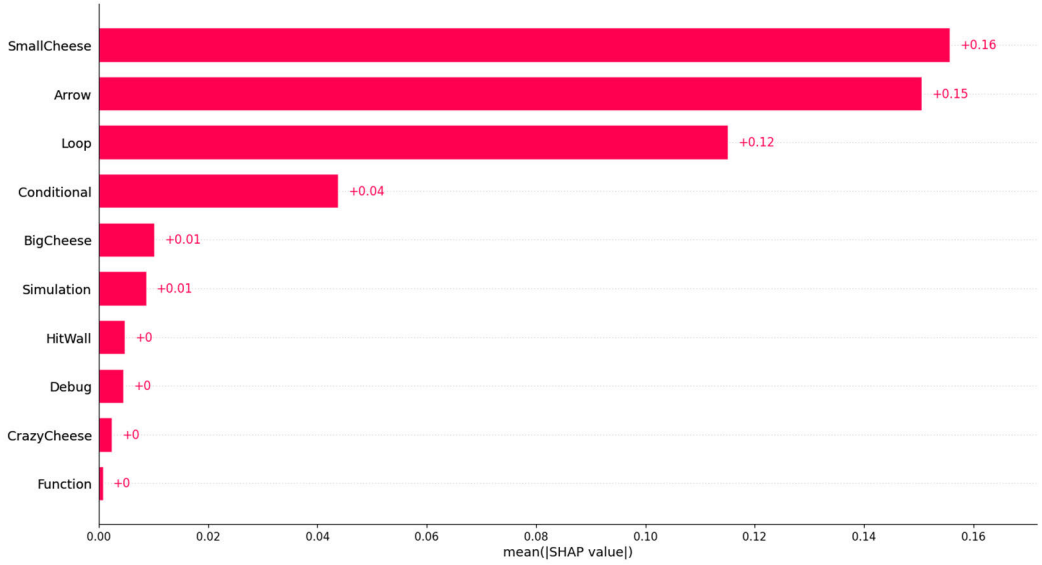


FIGURE 5. SHAP explainer global feature importance plot for the ANN.

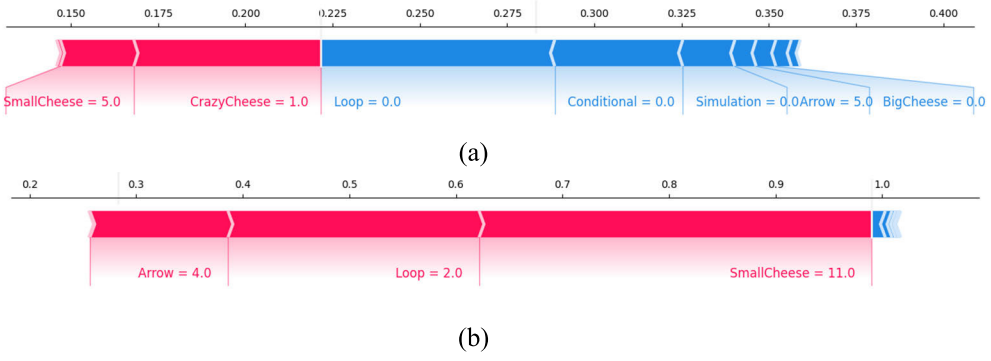


FIGURE 6. SHAP force plots for two examples with solution score labels: a) Low, and b) High.

SHAP values, ranging from −0.576 to 0.576, indicate the degree of impact (either decreasing or increasing) that each feature has on the model's outputs.

Each point on the Permutation SHAP plot (Fig. 8) represents the SHAP value for a specific feature in a single prediction instance, with the x-axis extending from negative to positive values to illustrate the extent of each feature's influence on decreasing or increasing the model output. The plot is color-coded, with blue points indicating a negative impact and red points a positive one, aiding in visually distinguishing the direction of each feature's impact. Features such as "SmallCheese," "Loop," and "Arrow" are prominently displayed with a broad spread of points along the x-axis, signifying their significant variability in influence. Notably,

TABLE 5. Feature importance scores from four different model explainers.

Feature	Kernel_SHAP	Permutation_SHAP	LIME	SHAP_Explainer
Arrow	0,967	0,968	0,672	0,967
Loop	0,741	0,739	1,000	0,739
Conditional	0,278	0,281	0,696	0,281
Function	0,011	0,005	0,080	0,005
Debug	0,029	0,029	0,171	0,029
Simulation	0,053	0,056	0,121	0,056
HitWall	0,028	0,030	0,097	0,030
SmallCheese	1,000	1,000	0,809	1,000
CrazyCheese	0,019	0,016	0,064	0,015
BigCheese	0,063	0,065	0,226	0,065

''SmallCheese'' exhibits a high concentration of red points towards the positive side, indicating a consistently strong and positive effect on the model's predictions.

Interestingly, in both plots, features ''SmallCheese,'' ''Loop,'' and ''Arrow'' emerge as the top three influencers, with ''SmallCheese'' being paramount in driving the predictions. Conversely, features such as ''Debug,'' ''Function,'' and ''Simulation,'' which are theoretically significant predictors, appear to be underutilized in predictions. This discrepancy reveals a potential oversight in model explanation training, where features critical to decision-making are neglected, while others that may not be inherently linked to core theoretical knowledge are overemphasized, possibly learning spurious correlations.

TABLE 6. Performance of the simulated predictions.

Models	Accuracy	Precision		Recall		F1-score	
		Low	High	Low	High	Low	High
KBANN	0.90	0.92	0.87	0.94	0.81	0.93	0.84
simulated SHAP-based predictions (Kernel)	0.79	0.92	0.63	0.77	0.85	0.84	0.72
simulated SHAP-based predictions (Permutation)	0.90	0.92	0.87	0.94	0.81	0.93	0.84
simulated LIME-based predictions	0.88	0.88	0.88	0.96	0.71	0.91	0.78

According to the global LIME feature importance plot (Fig. 9), the feature ''Loop'' emerges as the predominant influencer, followed closely by ''Conditional'' and ''SmallCheese,'' highlighting their critical roles in the model's decision-making mechanism. This contrasts with

TABLE 7. Correlation between the simulated predictions and KBANN predictions.

Models	Pearson Correlation
KBANN vs. Kernel SHAP	0,196642
KBANN vs. Premutation SHAP	0,19844
KBANN vs. LIME	0,447216

the Permutation SHAP plot (Fig. 8), where features like ''SmallCheese,'' ''Loop,'' and ''Arrow'' display significant variability in influence, with ''SmallCheese'' consistently showing a strong positive effect on the model's predictions. Despite these differences, both LIME and SHAP analyses identify ''SmallCheese,'' ''Loop,'' and ''Arrow'' as top influencers, indicating some alignment in feature importance assessment. However, LIME emphasizes ''Loop'' and ''Conditional'' more strongly, while SHAP provides a nuanced view with ''SmallCheese'' as a prominent predictor. This coherence, alongside discrepancies, suggests that while both methods agree on certain influential features, they reveal differences in how features like ''HitWall'' and ''Conditional'' are weighted. Additionally, features such as ''Debug,'' ''Function,'' and ''Simulation'' are underutilized in both analyses, indicating a potential oversight in the model's feature utilization strategy and the necessity to re-evaluate the model's explanation training process to ensure theoretically significant predictors are adequately recognized.

The bar chart (Fig. 10) displays the feature importance rankings from four distinct analytical models: KBANN, Kernel SHAP, Permutation SHAP, and LIME. KBANN distinguishes itself by incorporating domain knowledge and theoretical frameworks into its analysis, assigning significant importance to features recognized as critical within the CT framework, including ''Loop,'' ''Conditional,'' and ''Arrow'' for CT concepts, and ''Debug,'' ''Simulation,'' ''Function,'' and ''HitWall'' for CT skills. This focused approach underscores these features' pivotal role in evaluating task performance based on the mastery of both CT skills and concepts.

Contrastingly, Kernel and Permutation SHAP methods, while acknowledging these features, generally attribute lower importance to them, suggesting a discrepancy in how they assess their impact on model outputs. LIME exhibits considerable variation in its feature importance assessments and does not consistently align with the structured knowledge that KBANN integrates. This variability highlights a crucial issue with post-hoc explanation approaches. Our research shows that these methods may fail to correctly interpret the underlying reasons for predictions made by the reference KBANN model. Instead of accurately reflecting the decision-making process based on the model's theoretical rules, they frequently misattribute the importance to less relevant features. This misalignment challenges the reliability of their explanations and reveals their limitations in accurately conveying why the original model arrived at specific conclusions.

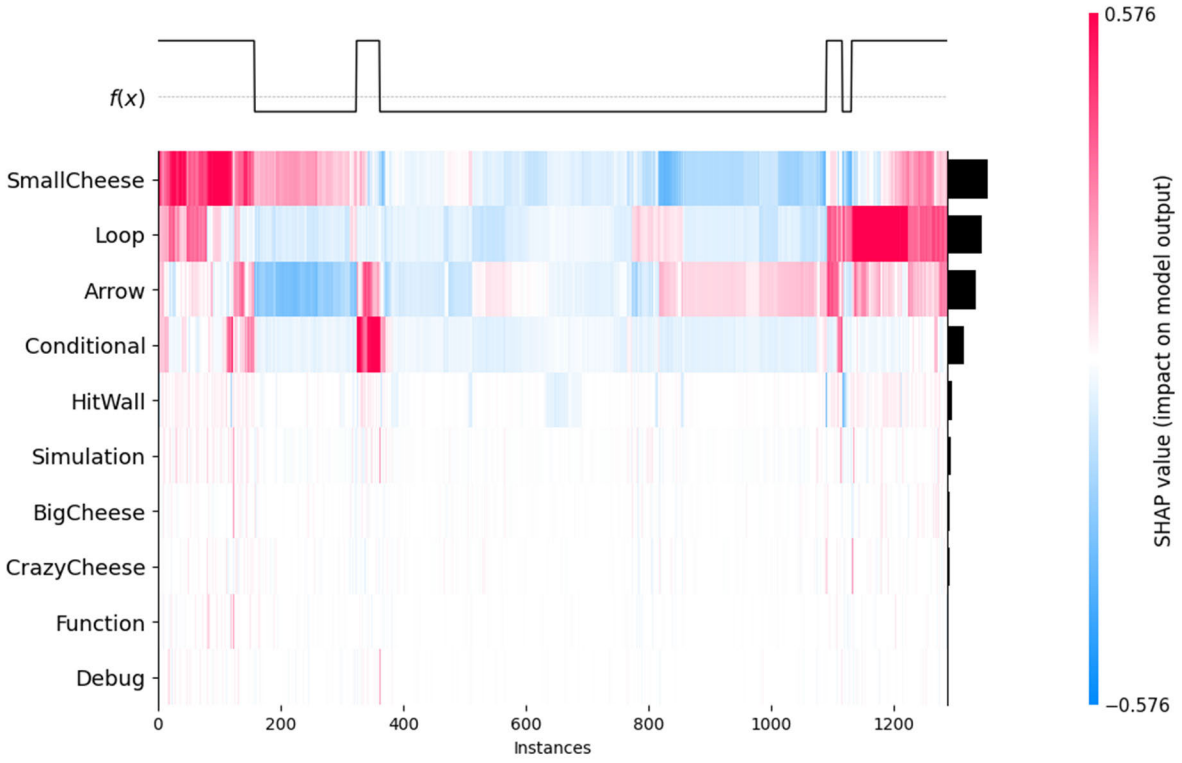


FIGURE 7. Global explanation of the simulated SHAP-based predictions (Kernel).

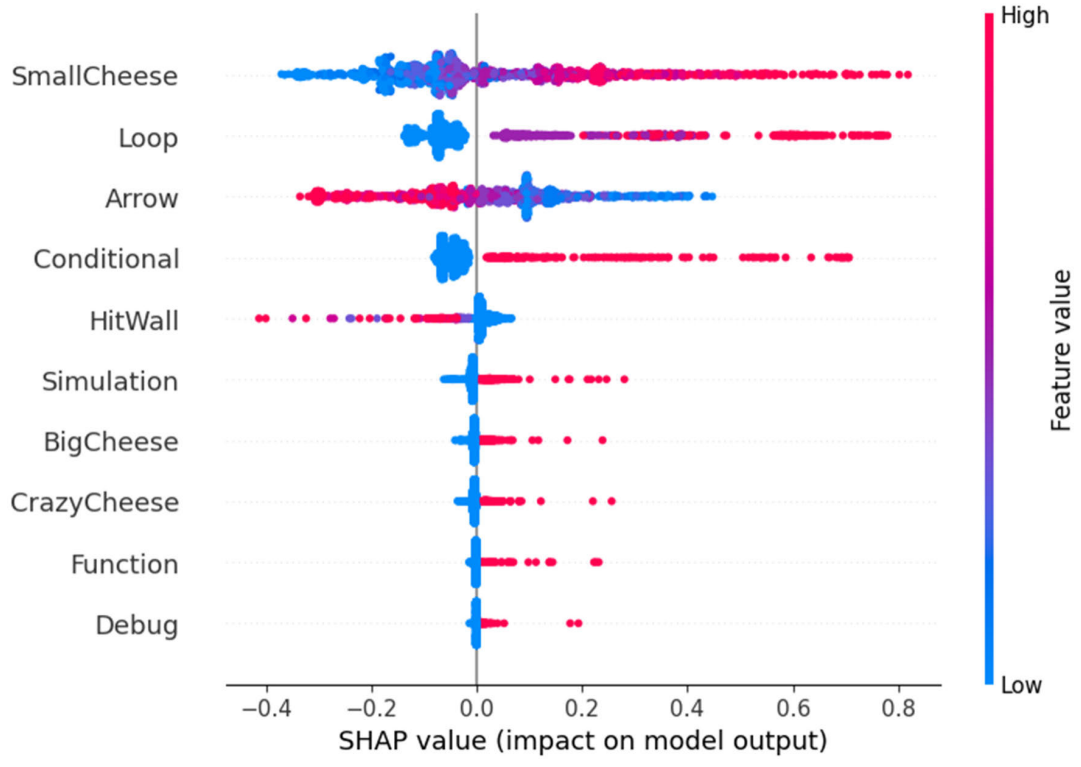


FIGURE 8. Global explanation of the simulated permutation SHAP-based predictions.

Given these findings, both researchers and stakeholders should approach the use of post-hoc explanation models with caution, especially in educational and other sensitive contexts where domain-specific knowledge is paramount. It is vital to

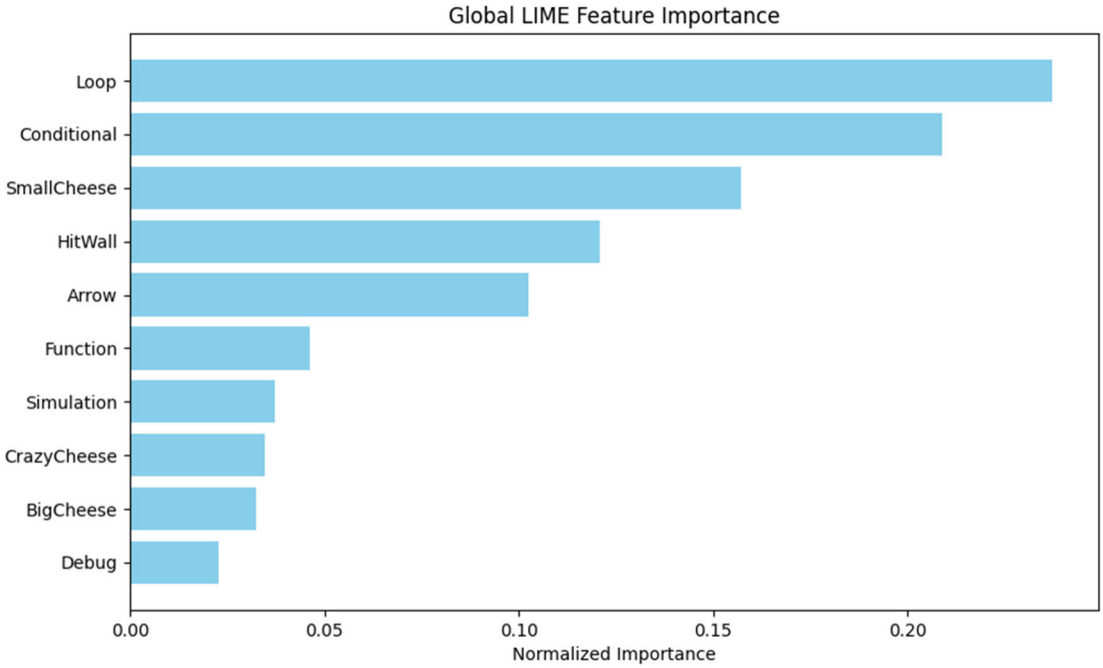


FIGURE 9. Global explanation of the simulated LIME-based predictions.

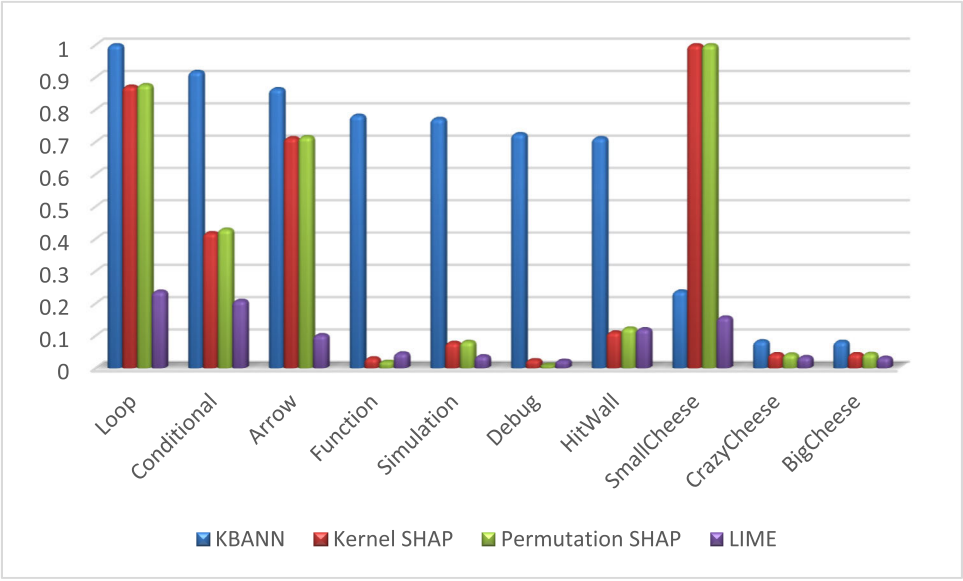


FIGURE 10. Comparing the feature importance deduced from SHAP and LIME against KBANN rules.

choose models that not only offer explanatory insight but also conform to relevant theoretical frameworks, ensuring that the interpretations are not only applicable but also grounded in established knowledge. This approach enhances the reliability and accuracy of AI systems in providing interpretations that genuinely reflect the predictive model's decision-making process.

3) COMPUTATION TIME

Table 8 shows the computation times for KBANN rule extraction, Kernel SHAP, Permutation SHAP, and LIME methods.

TABLE 8. Time cost of KBANN, Kernel SHAP, permutation SHAP, LIME.

Method	KBANN	Kernel SHAP	Permutation SHAP	LIME
Time Cost	18.44	573.51	340.1	133.07

KBANN is the fastest, taking only 18.44 seconds, demonstrating more efficient computation time and highlighting its efficiency over other methods. LIME is the second fastest at 133.07 seconds, while Kernel SHAP is the slowest at 573.51 seconds.

V. DISCUSSION AND CONCLUSION

This comparative study investigating the use of SHAP and LIME for interpretability in educational AI underscores significant challenges and complexities in employing these post-hoc explanation methods. The analysis of two experiments reveals key insights into the fidelity and reliability of these methods when aligned against the robust, theory-driven KBANN model.

The findings from the first experiment, comparing the KBANN and a conventional ANN in terms of generalizability and interpretability, serve as a basis for a nuanced critique of these methods. Firstly, the similar performance metrics (accuracy, precision, recall, F1-score) of both models may indicate that both models are equally effective. However, this similarity in metrics hides deeper issues related to how well we can interpret the models, understand their inner mechanisms, and ensure they follow educational principles. It is crucial to recognize that while ANN might achieve high performance, it relies significantly on potentially spurious correlations, as indicated by its emphasis on less causally relevant features like ''SmallCheese.'' For example, in the AutoThinking game, there is a strong correlation between small cheese consumption and the solution score. While the inclusion of the ''SmallCheese'' feature could offer some useful information, there is no causal relationship between this feature and the class label. Higher small cheese consumption does not necessarily indicate better CT concepts and skills, as many players use random strategies that can lead to high scores without employing core CT skills. In contrast, KBANN integrates structured educational knowledge, aligning more closely with causally relevant educational variables, thus providing a more robust basis for meaningful predictions within the educational context.

The post-hoc explanation methods—SHAP and LIME—show considerable variability in their interpretation of feature importance, which raises concerns about their reliability. For instance, the high importance scores assigned to features like ''Arrow'' and ''SmallCheese'' across various explainers used in the ANN model suggest a consistency that might seem reliable. However, the divergence in the importance of ''Conditional'' and ''Simulation'' across explainers reveals an underlying inconsistency. LIME's relatively higher scoring of these features compared to other methods could indicate its sensitivity to model-specific nuances, yet this also exemplifies the potential for different explanations to lead to conflicting interpretations about what features are truly driving predictions. This finding is in line with those reported by [28], [55], and [56] in that they highlight the fragility and variability of interpretation methods, leading to inconsistent and potentially misleading feature importance assessments.

The results from the second experiment provide a deeper insight into the challenges of post-hoc explanations. The investigation into the fidelity and reliability of these methods reveals significant disparities in their ability to mimic and interpret the decision-making processes of the KBANN when used as a reference model. First, the varied performance metrics of Kernel SHAP, LIME, and Permutation SHAP in terms of precision, recall, and F1-score relative to KBANN rules illustrate that even where numerical alignment appears (as seen with Permutation SHAP), it does not necessarily indicate a true conceptual alignment with the reference model's decision-making processes. The correlation analysis, showing relatively low Pearson correlations for all models, further supports this view. Even the highest correlation achieved by LIME (0.447216) falls short of demonstrating a strong alignment, suggesting that these post-hoc methods may capture some influential features but ultimately fail to replicate the reasoning of the reference model. This finding is consistent with [28] and [33], who observed that SHAP was less effective at replicating the original model's behavior, indicating potential fidelity issues.

The global feature importance analysis underscores this point more starkly. The focus on features such as ''SmallCheese,'' ''Loop,'' and ''Arrow'' by the post-hoc methods, especially with ''SmallCheese'' being overemphasized, illustrates a potential misalignment with the core theoretical constructs of the KBANN model (i.e., reference model). This overemphasis could be indicative of learning spurious correlations rather than meaningful, theoretically informed insights. On the other hand, KBANN's focused analysis recognizes and assigns high importance to features like ''Debug,'' ''Simulation,'' and ''Function'' which are critical for assessing mastery in computational thinking skills and concepts. These discrepancies in feature prioritization between the models highlight a crucial flaw in the training of post-hoc explainer methods, namely, the neglect of deeper, more causally relevant features in favour of more superficial or less relevant ones. Moreover, the variability in feature importance assessments, particularly noted with LIME, raises concerns about the consistency and reliability of these methods. While Kernel and Permutation SHAP methods generally attribute lower importance to key features recognized by KBANN, LIME's inconsistency and considerable variation in assessing feature importance further demonstrate the challenges in aligning post-hoc explanations with structured knowledge embedded in models like KBANN. Additionally, the computation time analysis reveals that KBANN achieves its results much more efficiently, further emphasizing its practical advantages over the other methods.

In conclusion, the comparative study highlights significant limitations in the current application of SHAP and LIME for interpretability in AI for education. The divergence in feature importance and the low correlation with the theory-driven KBANN model underscore the necessity for more robust, knowledge-integrated interpretability methods, especially when it comes to ANN-based models. As [57] argues, explainable black box models in high-stakes decisions are problematic due to their lack of fidelity to the original computations, leading to potential misinterpretations and mistrust. Additionally, black box models often fail to incorporate critical external information, increasing the risk of significant errors. Instead, developing inherently interpretable

models can prevent the perpetuation of flawed practices and ensure more trustworthy and transparent decision-making processes. These findings stress the importance of advancing interpretability techniques, based on neural-symbolic computing methods [12], [36], [37], [58], that align closely with domain-specific knowledge to provide meaningful, reliable insights. However, when deploying these models in real-world educational settings, additional challenges may arise, such as the scalability of these methods across diverse educational contexts and the need for continuous updates to the symbolic knowledge base to reflect evolving educational standards. Ensuring that the models remain interpretable and effective in dynamic, real-world environments will require ongoing refinement and adaptation.

A. LIMITATIONS AND FUTURE WORKS

This study has several limitations. First, the analysis was based on a single case study, which may restrict the generalizability of our findings. Future research should incorporate a wider range of educational case studies to derive more robust and comprehensive conclusions. Additionally, integrating symbolic knowledge can be challenging due to the need for relevant background knowledge. Preparing symbolic knowledge requires access to appropriate training data and domain understanding, which may occasionally pose difficulties depending on the context. The study also lacks user evaluations of the different explanations, which are crucial for understanding their effectiveness. Future work should include presenting explanations to users for feedback, which will help refine these methods. Furthermore, certain model assumptions in the experimental setting, such as the choice of network architecture, might affect the findings and could influence the broader applicability of the results. Addressing these assumptions in future work could help refine the models and provide more accurate interpretations. Lastly, exploring and comparing a diverse range of post-hoc models could provide deeper insights, highlighting their strengths and weaknesses and advancing more robust interpretability methods in AI for education.

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